

Movement MRU

MRU: Uniform Rectilinear Motion

The motion we have studied so far is called MRU: Uniform Rectilinear Motion. What does this mean?

- **Rectilinear:** it moves in a straight line. In this type of motion there are no curves.
- **Uniform:** the speed is constant. In other words, there is no acceleration.
 - It is important to understand that when there are several sections, this means that the speed is constant in each section. However, the speed CAN change from one section to another. We consider each section independently and we do not take into account the change between sections, so we ignore acceleration. This is a simplification, but it is very useful when we are starting to understand motion.

Equations of MRU motion

It is essential to learn the following equations very well. As you can see, they are the same equations we have already studied, even if they may look different.

- d or s or e : distance travelled (normally in m or km).
- x : position at a given moment (normally in m or km).

! Very important

$$\begin{aligned}\Delta s &= s_f - s_i \\ \Delta x &= x_f - x_i \\ \Delta t &= t_f - t_i \\ v &= \frac{s}{t} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}\end{aligned}$$

From these equations, you must also know how to rearrange the formulas:

! Very important: Rearranging equations

$$\begin{aligned}s &= \Delta x = v \cdot t \\ t &= \Delta t = \frac{s}{v} \\ x_f &= x_i + s = x_i + v \cdot t\end{aligned}$$